

INDUSTRIAL PRODUCT DATASHEET

reComputer Super J4012 - Advanced Edge AI Computer with NVIDIA® Jetson™ Orin™ NX 16GB in Super Mode

reComputer Super J4012 advanced edge AI computer with NVIDIA Jetson Orin NX 16GB in Super Mode, delivering up to 157 TOPS for vision AI, generative AI, and robotics.

SKU

114110314

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PRICE

\$1449.00(Excl. VAT)



Overview



The reComputer Super J4012 is an advanced edge AI computer featuring NVIDIA Jetson Orin NX 16GB in Super Mode, delivering up to 157 TOPS with configurable 10W–40W power modes, robust thermal design, dual GbE, LTE, 4x MIPI-CSI, and pre-installed JetPack 6.2 on a 128GB NVMe SSD. It supports rapid development for vision AI, generative AI, robotics, and multi-camera deployments through tools like reComputer AI Lab, Seeed Jetson Develop Tool, and Jetson-example.

Key Features

- **Super Boost AI Performance:** Delivers up to 157 TOPS in MAXN Super Mode, ideal for demanding applications like vision AI, robotics, and generative AI.
- **Energy-Efficient Operation:** Configurable power modes from 10W to 40W allow for optimized performance and energy consumption.
- **Robust Thermal Design:** Operates reliably in temperatures from -20°C to 60°C at 40W, and up to 65°C at 25W.
- **Rich interfaces:** Features 2x RJ45, SIM card, 4x USB 3.2, HDMI 2.1, and CAN. Also including M.2 Key E/M, Mini-PCIe, 4x Mipi-CSI for connectivity and cameras.
- **Ready for Deployment:** Pre-installed with JetPack 6.2 ([Supports JP7.2](#)) and a 128GB NVMe SSD, supporting AI frameworks like NVIDIA Isaac, Hugging Face, and ROS2/1, enabling rapid development of various AI applications.
- Memory Bandwidth Upgrade
- Higher Throughput
- Improved higher frame rate processing
- Enhanced Power Management
- Reduced Inference Latency

Built for developers, the reComputer Super J4012 is an enhanced upgrade over the [reComputer J4012](#). It takes edge AI development further with key enhancements: dual GbE (RJ45) ports, LTE, 4x Mipi-CSI for Camera, and full support for MAXN Mode.

Product Gallery



reComputer Super J4012

Jetson Orin NX 16GB (MAXN)

157 TOPS
AI performance

Wide range operating temperature under MAIN mode from -20°C to 60°C (Max to 85°C under 25W mode)

Open Source Hardware

Power Efficient **10W - 40W**

130mm x 120mm x 66mm

4x USB 3.2 Type-A (5Gbps)

pre-ins-39V

4x CSI

1x HDMI 2.1

mini PCIe for LTE 4G

2x RS-485 CbE (1Gbps) (1Gbps)

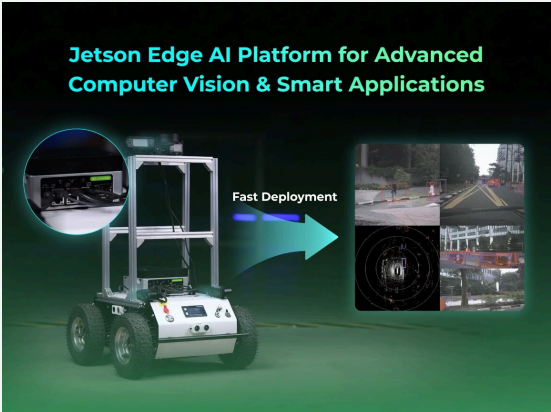
4x Antenna hole

WiFi/Bluetooth M.2 KEY E

Robotics

Generative AI

Smart Transportation



reComputer J4012 SERIES

SUPER vs ROBOTICS

Two edge AI systems. Two deployment priorities.

reComputer Super J4012

reComputer Robotics J4012

KEY DIFFERENCES

- 4x USB 3.2 - 4x MIP CSI
- 2x GbE - 1x CAN
- 12-19V DC - 40-Pin Header

BEST FOR

LLM & VLM at the Edge
Multi-Camera Vision
High-Performance AI

KEY DIFFERENCES

- 6x USB 3.2 - Optional GMSL2
- 2x GbE - 5x CAN
- 12-54V DC - UART / PC

BEST FOR

AMRs & Humanoids
Robotic Arms
Sensor Fusion & SLAM





Specification

	reComputer Super J4011	reComputer Super J4012
Applications Processor(AP)	NVIDIA Orin™ NX 8GB	NVIDIA Orin™ NX 16GB
AI Performance	Orin NX 8GB – 117 TOPS(MAXN)	Orin NX 16GB – 157 TOPS(MAXN)
GPU	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores	
GPU Max Frequency	1173 MHz	
CPU	6-core NVIDIA Arm® Cortex A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3	8-core NVIDIA Arm® Cortex A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3

CPU Max Frequency	2.0 GHz	
DL Accelerator	1 x NVDLA v2.0	2 x NVDLA v2.0
DL Max Frequency	1.23 GHz	
Vision Accelerator	1 x PVA v2.0	
Memory	8GB 128-bit wide LPDDR5 DRAM 102.4 GB/s	16GB 128-bit wide LPDDR5 DRAM 102.4GB/s
Power	10W - 15W - 25W - 40W	10W - 15W - 25W -40W
Video Encode	1x 4K60 (H.265) 3x 4K30 (H.265) 6x 1080p60 (H.265) 12x 1080p30 (H.265)	
Video Decode	1x 8K30 (H.265) 2x 4K60 (H.265) 4x 4K30 (H.265) 9x 1080p60 (H.265) 18x 1080p30 (H.265)	
CSI camera	Up to 4 cameras (8 via virtual channels***) 8 lanes MIPI CSI-2 D-PHY 2.1 (up to 20Gbps)	
Mechanical	69.6mm x 45mm 260-pin SO-DIMM connector	

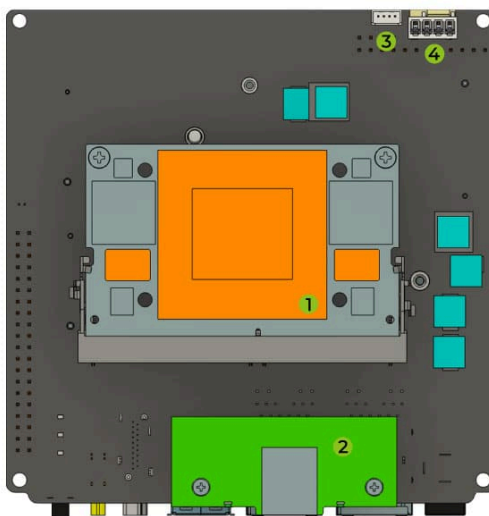
Carrier Board		
Storage	1x M.2 KEY M PCIe (M.2 NVMe 2280 SSD 128G included)	
Networking	M.2 KEY E	1x M.2 Key E for WiFi/Bluetooth module (Module not included)
	Mini PCIe	1x mini-PCIe for LTE 4G module (Module not included)
	Ethernet	2x RJ45 Gigabit Ethernet
I/O	USB	4x USB 3.2 Type-A (5Gbps); 1x USB 2.0 Type-C (Device Mode/Debug);
	Camera	4x mipi CSI(2-lane 15-Pin)
	CAN	1 x CAN(4-Pin Header)
	Display	1x HDMI 2.1
	Fan	1x 4-Pin 1.25mm Fan Connector (5V PWM) 1x 4-Pin 2.54mm Fan Connector (12V PWM)
	Extension Port	1x 40-Pin extension header 1x 12-Pin control and UART header
	RTC	1x RTC 2-pin 1x RTC Socket
	LED	2x LED(PWR and ACT)
	Button	1x PWR 1x RESET
	Switch	1x REC
	Antenna hole	4x Antenna Hole
Power	12-19V 5525 Barrel DC Jack	
Jetpack Version	Jetpack 6.2	
Mechanical	Dimensions (W x D x H)	130mm x 120mm x 66mm
	Weight	1110g
	Installation	Desk, Wall-mounting
Operating Temperature	-20°C~65°C(25W Mode) -20°C~60°C(MAXN Mode)	
Warranty	2 Years	
Certification	RoHS,REACH,CE,FCC,UKCA, KC	

Hardware Overview



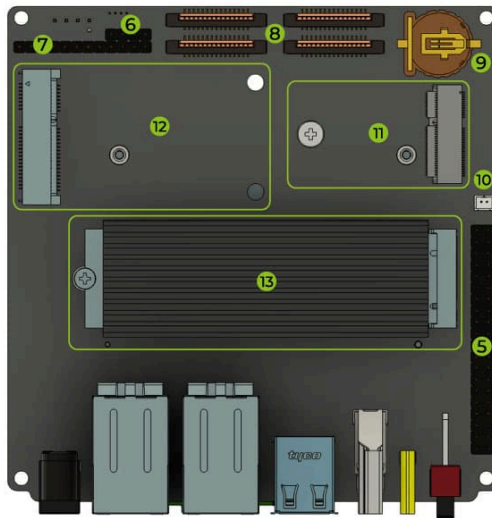
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|------------------|---------------------|-------------------------|
| ① DIP Switch | ④ 4x Type A USB 3.2 | ⑦ SIM Card Slot |
| ② Type C USB 2.0 | ⑤ 2x Pinhole Button | ⑧ 2x RJ45 Ethernet Port |
| ③ HDMI 2.1 | ⑥ 2x LED Indicator | ⑨ Barrel DC Jack 5525 |

Top View



- | |
|---------------------------------|
| ① Nvidia Jetson Orin Module |
| ② Extension Board |
| ③ 4-Pin Fan Connector (5V PWM) |
| ④ 4-Pin Fan Connector (12V PWM) |

Bottom View



- 5 40-Pin Extension Header
- 6 CAN Header
- 7 12-Pin control and UART header
- 8 4x 15-Pin MIPI CSI
- 9 RTC Battery Socket(CR1220)
- 10 RTC 2-Pin Header(3V)
- 11 M.2 Key E Slot
- 12 Mini PCIe Slot
- 13 M.2 Key M Slot(with SSD)

Applications

- **NVIDIA Jetson for Beginners:** Whether you're an experienced developer or a complete beginner for embedded AI/GPU inference, use reComputer Super AI box as your first AI workstation, together diving into NVIDIA Jetson for Beginners — your fast track to vision AI, Generative AI, and robotics development!
- **From Zero to Real Projects:** Master Linux basics on the Jetson environment to build an end-to-end AI NVR/AMR.
- **Hardware Integration:** Tailored for reComputer Jetson devices for a seamless, out-of-the-box experience.
- **Coverage of Cutting-Edge Domains:** Deep dive into CV, GenAI, ROS, agentic AI, and TensorRT optimization.
- **Toolchain Mastery:** Hands-on tutorials with NVIDIA JetPack SDK, CUDA, DeepStream, PyTorch, and TensorFlow.
- **reComputer AI Lab:** To improve the developer experience, Seeed offers reComputer AI Lab — a platform that helps you deploy AI applications with one-click tools and ready-to-use resources. It includes optimized edge AI model demos for CV, LLM, VLM, interactive tutorials, deployment tools, containerized applications, and community projects to speed up development on Jetson.
- **Seeed Jetson Develop Tool:** Seeed Jetson Develop Tool is a desktop client for all developers who are going to use reComputer Jetson device for the very first time. It brings the common bring-up and development workflow into one GUI: firmware flashing, first-boot initialization, device diagnostics, remote development, app deployment, skill-based automation, OTA updates, and community resources. All we provide here is to help you to get started with Seeed reComputer Jetson hardware much more easily and seamlessly.
- **Jetson-example:** Ready to build a solution for the real world? We have Jetson-example prepared for you! It offers one-line deployable projects compatible with generative AI large models, including LeRobot, Ollama, Qwen, etc; with computer vision such as Ultralytics YOLO26. We have configured all environment setups for you to provide a single command deployment of projects, easy to try out/verify if the trending model performance is suitable for your GPU needs and application scenarios.
- **Multi-Camera Vision AI:** Powered by the reComputer Super AI box, the system supports up to 4x USB cameras or MIPI-CSI cameras to create a 360° panoramic vision network. Leveraging the advanced spatial foundation model, it processes multiple real-time video streams simultaneously to deliver accurate depth perception, comprehensive environmental awareness, and real-time blind-spot elimination for safer autonomous operation.

Part List

Jetson Orin™ NX 16GB/NX 8GB /Nano 8GB/Nano 4GB module	x 1
Seeed Carrier Board (reComputer Super J401)	x 1
128GB NVMe SSD	x 1
Aluminum Case and Heatsink with Fan	x 1
USB Cable; Type A to Type C	x 1
User Manual	x 1

Compliance & Logistics



HSCODE	8471504090
USHSCODE	8517180050
UPC	
EUHSCODE	8471707000
COO	CHINA

FCC Requirement

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) this device may not cause harmful interference, and

(2) this device must accept any interference received, including interference that may cause undesired operation.

15.21

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy.

If not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Resource Links

- <https://www.seeedstudio.com/reComputer-Super-J4012-p-6443.html>
- [User Manual&Datasheet](#)
- [Temperature Test Report](#)
- [Schematic](#)
- [3D File](#)
- [Mechanical Document-reComputer Super](#)
- [Mechanical Document-reComputer Super PCBA](#)
- [Seeed Nvidia Jetson Success Cases](#)
- [Seeed Jetson One Pager](#)
- [Flash BSP for Jetson](#)

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